A Finite-State Decision-Making Framework for Autonomous Navigation in Urban Scenarios

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***Abstract*—This paper presents the design, implementation, and evaluation of a decision-making module for autonomous navigation in urban environments, based on two coordinated finite-state machines (FSMs) and integrated into a pre-existing modular autonomous driving architecture. The proposed decision-making module acts as a high-level decision layer operating between the perception and control modules. It combines a Longitudinal Decision FSM and a Lateral Maneuver FSM under an explicit Safety Priority and Conflict Resolution Logic. Its objective is to select safe, interpretable, and easily adjustable driving behaviors. The system is implemented in Python and validated in the CARLA simulator using representative urban scenarios. The obtained results show the ability of the proposed approach to generate traceable, coherent, and safety-oriented decisions, consolidating it as an integrable solution within the architecture.**

***Index Terms*—Autonomous driving, decision-making, finite-state machine, modular architecture, urban navigation, CARLA.**

I. Introduction

Autonomous driving is key to improving transport safety and efficiency. However, achieving safe and robust autonomy requires overcoming significant technical challenges due to the inherent complexity of urban environments. These scenarios are highly dynamic and uncertain, characterized by continuous interaction with heterogeneous agents (pedestrians, cyclists, vehicles at varying velocities) and the need to comply with traffic regulations [1]. In this context, ensuring safety has become the primary bottleneck in the industry.

This work focuses on the local planning module within a modular architecture, specifically addressing the decision-making layer. This module acts as a bridge between scene understanding (perception) and physical execution (control). Its function is to process the vehicle state, the planned route, and environmental constraints to select safe, efficient high-level behaviors consistent with traffic rules.

This project is developed within the research framework of the RobeSafe group, which possesses an autonomous driving platform and a consolidated modular architecture including advanced perception, localization, and map-monitoring systems [2]. Based on this foundation, a decision-making module is proposed based on two coordinated finite-state machines (FSMs), structured through explicit longitudinal and lateral states, transitions, and safety-priority rules. The proposed design separates the Longitudinal Decision FSM from the Lateral Maneuver FSM, improving traceability while preserving flexibility for overtaking and route-recovery behaviors.

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II. Related Work

A. Autonomous Driving Architectures

Driving architectures can be grouped into: End-to-End (E2E) and modular approaches [3], [4]. E2E architectures utilize deep learning models to directly map sensory inputs to control actions within a single global network. They simplify system structure and enable joint optimization. However, they exhibit lower interpretability and high training-data dependency, which can limit their application in safety-critical systems [3], [5]. Modular architectures divide the system into separate functional blocks, commonly associated with perception, localization, planning, and control [4]. This approach allows each component to be developed, validated, and modified in isolation, facilitating system traceability and error identification [6].

In this work, a modular architecture is adopted because it enables the integration of the decision-making module into the pre-existing architecture and facilitates the traceability, validation, and independent tuning of its decisions.

B. Decision-Making Techniques for Autonomous Driving

The decision-making system must generate safe and reasonable driving behaviors considering environmental information, the motion of other traffic participants, the vehicle state, and traffic constraints [7], [8]. According to the classification in [7], these techniques are categorized into:

1) Classical Decision-Making Methods: These comprise rule-based methods, optimization methods, and probabilistic methods [7]. Rule-based methods operate under strict traffic regulations. Notable examples include FSMs and Behavior Trees (BTs), where the latter mitigates the state explosion of the former by structuring logic hierarchically [9]. On the other hand, optimization-based and game-theoretic methods balance safety and efficiency at the expense of higher computational load [10]. Finally, probabilistic methods explicitly incorporate environmental uncertainty.

2) Learning-Based Decision-Making Methods: These utilize artificial intelligence techniques to learn behaviors from data or environmental interaction, excelling in complex scenarios [7]. While classical statistical methods require well-defined variables, Deep Learning processes raw sensory data, typically integrating into E2E architectures. Alternatively, Imitation Learning infers control actions by mimicking expert demonstrations [8], and Reinforcement Learning optimizes decisions through a reward system, though it may suffer from stability and overfitting issues.

Considering the requirements of a safety-oriented modular stack, this work adopts an FSM-based approach, since it provides deterministic and interpretable behavior, low computational overhead, and explicit mechanisms to enforce safety and regulatory compliance.

C. Finite-State Machines for Behavioral Decision-Making

FSMs model discrete behaviors through explicit states and transition conditions and have been widely used in autonomous driving systems since early urban navigation platforms such as Boss [11]. Their selection for the decision module relies on three core pillars: they provide an interpretable and adjustable logic, are computationally lighter and more direct than alternative approaches, and allow enforcing strict regulatory frameworks where collision avoidance always supersedes traffic fluency. Furthermore, they do not require complex hierarchical control structures nor depend on training datasets, facilitating their integration across diverse architectures. Several works have applied FSM-based strategies to different autonomous driving tasks. Wang et al. [12] proposed a hierarchical decision-making framework composed of three FSM layers responsible for scenario classification, behavior evaluation, and action selection. Zheng et al. [13] combined Artificial Potential Fields and FSMs to identify vehicle states and trigger emergency braking or evasive lane-changing maneuvers. More recently, the WATonoBus platform [14] employed an FSM as the core of a modular decision-making architecture, managing operational states including normal driving, intersection handling, merging, pullover maneuvers, and emergency stopping.

Compared with these approaches, the proposed decision-making module combines two coordinated FSMs: a Longitudinal Decision FSM for constraint-aware velocity selection and a Lateral Maneuver FSM for overtaking and return maneuvers.

D. Contributions

The main contributions of this work are:

1) Design and implementation of a dual-FSM decision-making module for urban autonomous navigation.
2) Definition of a Safety Priority and Conflict Resolution Logic to handle simultaneous constraints.
3) Integration of the proposed decision-making module into a pre-existing modular autonomous driving architecture.
4) Validation in CARLA using a selected subset of Bench2Drive routes covering representative urban driving scenarios.

III. Proposed Decision-Making Module

A. System Architecture

The proposed decision-making module is integrated into the pre-existing modular autonomous navigation architecture as a replacement for its original decision-making component [2]. In this setup, perception, localization, and global planning modules provide a structured environmental representation, while the control module executes longitudinal and lateral references. The decision-making module is positioned between these components as shown in Figure 1.

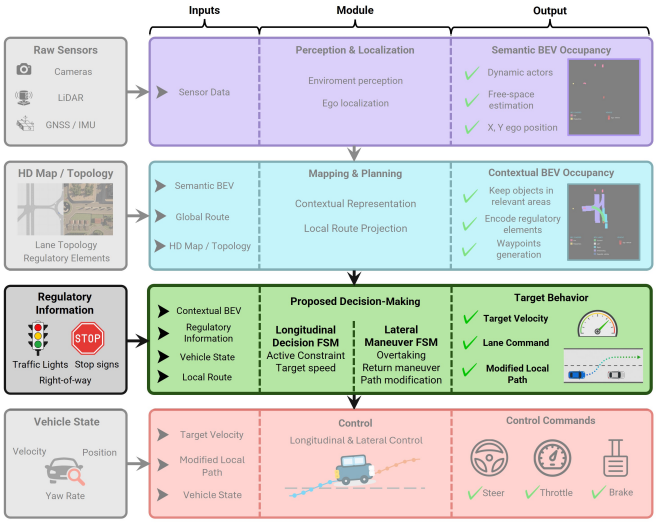


Fig. 1. Simplified architecture of the autonomous navigation system, adapted from [2]. The highlighted decision-making block corresponds to the proposed FSM-based module, composed of the Longitudinal Decision FSM and the Lateral Maneuver FSM.

The internal architecture of the proposed decision-making module is divided into two synchronized components:

- **Longitudinal Decision FSM:** Processes the contextual bird's-eye view (BEV), regulatory information, vehicle state, and local route to estimate the active constraint and generate the target velocity.
- **Lateral Maneuver FSM:** Manages overtaking and return maneuvers by modifying the local route waypoints when a lane change is authorized.

B. Decision Module Inputs and Outputs

The decision-making module processes heterogeneous data originating from perception, localization, and global planning modules. It receives environmental and road-topology information through semantic and contextual BEV occupancy representations, together with regulatory information, vehicle state, and local route data. The module processes these inputs to extract distance metrics, active constraints, and lane availability.

The primary output is a target reference velocity sent to the vehicle controller. Furthermore, when the system authorizes a lane change, the local route waypoints are modified in real time, generating an updated local path.

Table I summarizes the detailed classification of input and output variables for the implemented decision module.

TABLE I
INPUTS AND OUTPUTS OF THE DECISION-MAKING MODULE

Type	Category	Description
Input	BEV Perception	Semantic segmentation (vehicles, pedestrians) and lane topology (current lane, adjacent lanes, opposite lane and intersection areas).
	Infrastructure	Traffic light state, stop-sign information and metric distance to the upcoming junction.
	Obstacles	Front obstacle detection and cross-traffic distance.
	Ego state	Ego velocity, ego localization and yaw rate.
	Route Planning	Global route and local projected path waypoints.
Output	Target Velocity	Longitudinal velocity reference sent to the vehicle controller.
	Lane Command	Discrete maneuver command: Keep-lane or lane-change command.
	Modified Path	Local route waypoints dynamically modified during lane changes.

C. Finite-State Machine Design

The decision logic of the proposed module is illustrated in Figure 2. Following the architecture described above, the proposed decision-making module is composed of two synchronized FSMs: the Longitudinal Decision FSM and the Lateral Maneuver FSM. The Longitudinal Decision FSM selects the target velocity according to the active environmental constraint, while the Lateral Maneuver FSM modifies the local path during overtaking and return maneuvers.

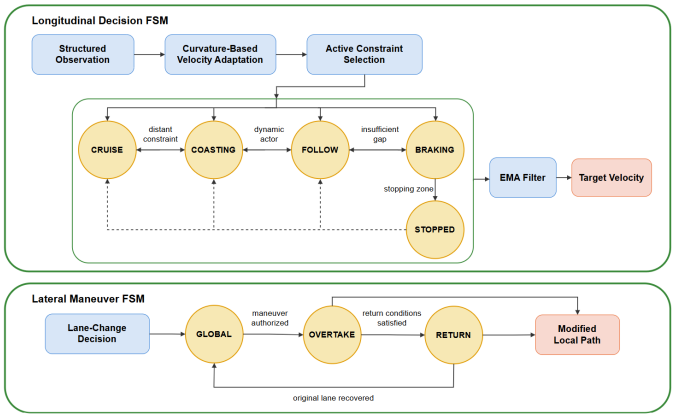


Fig. 2. Decision logic of the proposed module. The Longitudinal Decision FSM determines the target velocity according to the active constraint, while the Lateral Maneuver FSM handles overtaking and return maneuvers by modifying the local path.

1) Longitudinal Decision FSM: The Longitudinal Decision FSM is organized into four main stages: curvature-based velocity adaptation, active constraint selection, FSM state selection, and velocity filtering. These stages are re-evaluated at every decision cycle. Therefore, the transitions represented in Figure 2 should not be interpreted as a mandatory sequential chain, since the FSM may directly select the state required by the current situation.

Curvature-based velocity adaptation. Before evaluating the active state, the Longitudinal Decision FSM adapts the maximum admissible velocity to the local road geometry. The route curvature radius R is estimated from sets of three spaced waypoints using the circumcircle defined by the corresponding waypoint triangle. A smaller radius indicates a higher curvature in the evaluated segment. Once the curvature radius is obtained, vehicle dynamics are considered. To prevent loss of traction, the system caps lateral acceleration at a safe maximum threshold (a_{lat_max}). Under these conditions, and using the centripetal acceleration formula, the FSM output velocity is bounded according to Eq. (1).

$$v_{lim} = \sqrt{a_{lat_max} \cdot R} \tag{1}$$

Active constraint selection. After adapting the velocity to the road geometry, the Longitudinal Decision FSM processes the structured environmental observation to identify active constraints. These include preceding vehicles, pedestrians in the ego lane, active traffic lights, STOP signs, front obstacles, and cross-traffic in intersections. Each candidate constraint is defined by a distance and a category. When multiple candidates appear simultaneously, the closest one is selected as the active constraint. The required braking distance is computed according to Eq. (2):

$$d_{req} = \frac{v^2}{2a_{dec}} + d_{stop} + 1.2v \tag{2}$$

where v is the current ego velocity, a_{dec} is the configured maximum deceleration, and d_{stop} is the safety stopping distance associated with the constraint type. This distance adapts to the active candidate. STOP signs use a specific approach distance, cross-traffic relies on an independent safety margin, and front obstacles are treated as static blocking constraints.

FSM state selection. Based on the active constraint, the available distance, and the required braking distance, the Longitudinal Decision FSM selects one of the five states illustrated in Figure 2. `CRUISE` represents free driving conditions. It is selected when no relevant constraint limits the vehicle motion or when the available distance allows the reference velocity to be maintained. `COASTING` provides an anticipatory response to distant constraints. It is activated when a relevant candidate is detected within the coasting range, but neither active following nor braking is yet required. `FOLLOW` is used when the active constraint is a dynamic actor located within the following range and the available distance remains greater than the required braking distance. For car-following, the FSM computes the target velocity through a proportional controller designed to maintain a safe time gap. Defining the distance error as $e(t) = d_{ahead}(t) - (d_{min} + T_{gap} \cdot v_{now}(t))$, the velocity is updated according to Eq. (3):

$$v_{target}(t) = v_{lead}(t) + K_p \cdot e(t) \tag{3}$$

where v_{lead} is the preceding vehicle's estimated velocity and K_p is an adaptive proportional gain.

`BRAKING` is selected when the available distance becomes lower than the required braking distance or when the active

candidate represents a stopping constraint. Such constraints include red traffic lights, STOP signs, static front obstacles, pedestrians requiring intervention, and cross-traffic conflicts. `STOPPED` represents the complete-stop condition. The state is reached when the ego vehicle enters the corresponding stopping zone and its velocity is close to zero. The vehicle remains stopped while the active constraint is still valid.

Velocity filtering and temporal memory. The raw target velocity computed by the FSM (v_{target}) is filtered through an asymmetric Exponential Moving Average (EMA), governed by Eq. (4). This filtering is used to dampen oscillations caused by sensory noise or intermittent detections.

$$v_{out}(t) = \alpha \cdot v_{out}(t-1) + (1-\alpha) \cdot v_{target}(t) \tag{4}$$

The inertia factor $\alpha \in [0, 1]$ adapts contextually, prioritizing smoothness during cruise mode ($\alpha = 0.8$) and nullifying inertia during emergencies ($\alpha = 0.0$) to enforce maximum braking. Additionally, to avoid unstable decisions triggered by momentary perception dropouts, the system incorporates temporal memories for traffic lights, obstacles, and cross-traffic. These memories keep constraints active for several cycles when detection drops briefly, preventing rapid chattering between states and preserving a more stable response against noise.

2) Lateral Maneuver FSM: The Lateral Maneuver FSM operates concurrently with the Longitudinal Decision FSM, as shown in the lower branch of Figure 2. Under normal conditions, the `GLOBAL` state preserves the original local route. When a lane-change maneuver is authorized, the Lateral Maneuver FSM transitions to `OVERTAKE` and progressively displaces the local waypoints toward the target lane. Once the obstacle or slower vehicle has been passed and the original lane can be safely recovered, the Lateral Maneuver FSM transitions to `RETURN`. The `GLOBAL` state is restored when the original route has been recovered.

When overtaking is authorized, local waypoints are adjusted through cosine-based interpolation according to the longitudinal progress $p \in [0, 1]$. The interpolation factor is computed using Eq. (5):

$$f(p) = \frac{1 - \cos(p \cdot \pi)}{2} \tag{5}$$

This factor modulates the lateral displacement, providing a smooth variation of the lateral offset at the beginning and end of the maneuver and avoiding abrupt waypoint displacements.

D. Safety Priority and Conflict Resolution Logic

The Safety Priority and Conflict Resolution Logic resolves simultaneous constraints by enforcing the following hierarchy:

- **Immediate Safety**: Reaction against pedestrians in the ego lane, nearby obstacles, and direct collision risks takes highest precedence. Under these circumstances, the system triggers `BRAKING` or `STOPPED` regardless of other available actions.
- **Regulatory Compliance**: Red or yellow traffic lights, STOP signs, and intersection right-of-way conflicts are handled as `STOPPED` constraints. While active, evasive lateral lane-change maneuvers are explicitly prohibited.
- **Longitudinal Safety**: When tracking a preceding vehicle or actor, the system maintains separation based on minimum distance and time gaps. If available distance falls below safe thresholds, state shifts from `FOLLOW` to `BRAKING`.
- **Lane-Change Authorization**: Lane changes are only evaluated if the adjacent lane exists and fulfills required safety margins. Lateral maneuvers are blocked if the active constraint involves traffic lights, STOP signs, or cross-traffic. Furthermore, overtaking is denied if the front obstacle lacks persistence, lead vehicle velocity is not sufficiently low, velocity gain is minimal, target lane is occupied, or route curvature renders initiation unsafe.

IV. Experimental Setup

A. Simulation Environment

The experiments are conducted in the CARLA simulator [15], an open-source platform oriented toward developing and evaluating autonomous driving systems in urban environments. Testing runs over standard simulator urban maps featuring complex topology, dynamic traffic, pedestrians, traffic lights, intersections, STOP signs, static obstacles, and diverse road layouts.

B. Evaluated Scenarios

To ensure a rigorous and representative assessment of urban traffic complexity, tests are structured using the standardized Bench2Drive benchmark [16]. Following this protocol, a subset of 44 routes designed to expose the ego vehicle to each of the 44 unique conflict topologies present in the benchmark was selected. Table II groups scenarios according to the evaluated interaction type, categorized adopting the benchmark's official driving ability taxonomy.

TABLE II
Evaluated Driving Scenarios

Driving Ability	Routes	Description / Key Conflicts
Merging	16	Main-road merges, parking lot exits, and interactions in unsignalized intersections with dense traffic.
Overtaking	9	Evasion of static obstacles, irregularly parked vehicles, and bicycles in adjacent lanes.
Emergency Brake	12	Critical braking against sudden intrusions (cut-ins), road accidents, and unexpectedly crossing pedestrians.
Traffic Signs	19	Strict adherence to traffic lights, STOP signs, and right-of-way management at regulated junctions.
Give Way	2	Specific interactions requiring yielding, such as approaching emergency vehicles.

Note: Certain routes feature overlapping conflicts assessing multiple abilities simultaneously, hence the total sum exceeds 44 scenarios.

C. Evaluation Metrics

In accordance with the validation framework proposed by [16], system performance is quantitatively evaluated utilizing its official standardized metrics.

The primary metric is the **Driving Score (DS)**, a composite score evaluating both navigation progress and vehicle safety. This metric is computed as the product of the following two components:

- **Route Completion (RC):** The percentage of the theoretical route completed autonomously by the ego vehicle before reaching its destination or triggering a fatal failure.
- **Infraction Penalty (IP):** A geometric multiplier (ranging within $[0.0, 1.0]$) penalizing gross progress based on infractions logged along the trajectory.

Additionally, the **Success Rate (SR)** is reported, indicating the percentage of routes successfully completed without relevant traffic infractions.

V. Results and Discussion

The experimental results are analyzed from three complementary perspectives: a general quantitative analysis summarizes the overall performance of the proposed module and examines the results across the evaluated driving abilities and observed infractions, a qualitative scenario-level analysis discusses representative behaviors of the decision-making system and the obtained results are contextualized against reference methods reported in the Bench2Drive benchmark.

A. Overall and Ability-Level Results

The proposed decision module achieved a mean DS of 71.92 over the 44-route Bench2Drive subset, driven by a high mean RC of 89.89% but tempered by a mean IP of 0.798, resulting in an SR of 43.18%. Of the 44 evaluated routes, 19 were completed perfectly, 18 were completed with infractions, and 7 resulted in critical failure.

TABLE III
Results by Bench2Drive Driving Ability

Ability	Routes	DS	RC (%)	SR (%)
Merging	16	65.19	91.51	25.00
Overtaking	9	70.45	82.39	66.67
Emergency Brake	12	85.63	100.00	58.33
Give Way	2	85.00	100.00	50.00
Traffic Signs	19	65.56	91.66	21.05

The ability-level analysis presented in Table III reveals clear differences across the evaluated interaction categories. The highest performance is achieved in Emergency Brake, with a DS of 85.63, and Give Way, with a DS of 85.00. Both categories achieve 100% RC, indicating that the module responds reliably to well-defined longitudinal constraints requiring braking or yielding. Overtaking achieves a DS of 70.45 and the highest SR among the evaluated abilities, reaching 66.67%. In contrast, Merging and Traffic Signs obtain the lowest SR values, with 25.00% and 21.05%, respectively. These categories concentrate dense interactive situations and multiple simultaneous constraints, where the module must combine longitudinal safety, regulatory compliance, and interaction with surrounding traffic.

To further investigate the causes of performance degradation, the distribution of the 36 infractions recorded across the 44 routes was analyzed. The most frequent failure category corresponds to vehicle collisions, with 16 occurrences. Detailed inspection indicates that a relevant proportion of these events occurs during highly interactive merging situations. In several scenarios, the ego vehicle initiates the maneuver while preserving the longitudinal safety criteria defined by the planner, but surrounding simulated agents maintain their trajectories and velocities without yielding sufficient space, resulting in rear or side impacts.

Five infractions were also recorded for lane departures and five for red-light violations, indicating that the management of traffic regulations remains sensitive to temporal instability in perception at complex intersections. In contrast, no pedestrian collisions and no minimum-velocity violations were recorded. These results indicate that the braking logic provides reliable responses to vulnerable road users, while the main remaining limitations are associated with vehicle-to-vehicle interaction and regulatory constraint stability in complex traffic situations.

B. Qualitative Analysis

The qualitative analysis examines representative behaviors observed during the evaluated scenarios. For regulatory constraints such as traffic lights and STOP signs, temporal memory mechanisms preserve active constraints during brief perception dropouts, avoiding rapid oscillations between `BRAKING` and `CRUISE` and preventing premature acceleration. Similarly, when interacting with preceding vehicles, the Longitudinal Decision FSM transitions between `COASTING`, `FOLLOW`, and `BRAKING` according to the available distance and braking requirements, progressively adapting the target velocity when necessary.



Fig. 3. System behavior during an overtaking maneuver.

Figure 3 illustrates a representative overtaking scenario. The ego vehicle encounters two slow-moving bicycles ahead in its lane, which become the active constraint for the decision-making module. Using BEV perception and lane topology information, the Lateral Maneuver FSM verifies lane availability and authorizes the maneuver, since all lanes on the multi-lane road carry traffic in the same direction. The Longitudinal

Decision FSM adapts the target velocity to the slower actors, while the Lateral Maneuver FSM modifies the local route to perform the lane change. This example illustrates the coordinated operation of the Longitudinal Decision FSM and the Lateral Maneuver FSM, allowing the vehicle to react to slow-moving actors without remaining unnecessarily blocked behind them.

C. Contextual Comparison

To contextualize the obtained results, Table IV compares the proposed module with reference methods reported by the Bench2Drive benchmark [16]. Since the proposed decision-making module was evaluated on a selected subset of 44 routes rather than the complete official 220-route benchmark, the comparison should be interpreted as contextual rather than as a direct ranking between methods.

TABLE IV
CONTEXTUAL COMPARISON WITH BENCH2DRIVE REFERENCE METHODS

Method	**DS**	**SR (%)**
AD-MLP	18.05	0.00
UniAD-Base	45.81	16.36
VAD	42.35	15.00
TCP-traj*	59.90	30.00
Think Twice	62.44	31.23
Drive Adapter	64.22	33.08
Proposed (FSM-based)	**71.92**	**43.18**

Note: Reference method scores are taken from the Bench2Drive benchmark [16]. Proposed method was evaluated on the selected 44-route subset.

These results suggest that an interpretable rule-based architecture can provide strong performance in structured urban scenarios. Nevertheless, the selected route subset was designed to cover the conflict types considered during development and validation of the proposed module. Therefore, the reported results do not demonstrate general superiority over learning-based approaches, which are evaluated under the complete benchmark protocol and may provide greater adaptability to previously unseen or more complex interaction patterns.

VI. CONCLUSIONS AND FUTURE WORK

This work presented the design, implementation, and evaluation of a dual-FSM decision-making module for autonomous navigation in urban scenarios. Integrated as an intermediate reasoning layer between perception and control, the proposed module manages high-level behaviors in a traceable and modular manner. By coordinating the Longitudinal Decision FSM and the Lateral Maneuver FSM under an explicit Safety Priority and Conflict Resolution Logic, the module achieves a balanced tradeoff between driving safety, regulatory compliance, and traffic fluency. Its finite-state structure enables vehicle behavior diagnosis and tuning through explicit rules, supporting the development of safe and traceable autonomous driving systems.

Future work will extend evaluation across the full 220 official Bench2Drive routes and explore evolving the FSM toward Behavior Trees to mitigate state explosion in highly complex settings. Furthermore, intention prediction models will be integrated to improve social interaction during dense merges, alongside investigating machine learning for adaptive parameter tuning. Ultimately, we aim to validate the module on physical platforms in real-world scenarios.

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